

Project G6: FracTrack — Deep Learning for X-Ray Fracture Analysis

- **Clinically motivated and well-grounded:** The literature review is thorough—you cover YOLO-based detection, hybrid HDR models, and foundation models (MedGemma), and correctly identify the gap that simpler architectures with transfer learning may match foundation-model performance at lower cost. The multi-task formulation (anatomy classification + abnormality detection) is a smart design choice. ConvNeXt is a strong backbone choice.
- **Clarify the task before coding:** You mention both classification (normal/abnormal) and detection (bounding boxes, YOLO-style). The MURA dataset provides *study-level* binary labels (normal/abnormal per study, each study has multiple images), *not* pixel-level or bounding-box annotations. FracAtlas has bounding boxes. Are you doing classification only (MURA) or also localization (FracAtlas)? These require different architectures and evaluation metrics. Commit to one primary task now.

Recommended actions before Milestone-2

- **Dataset verification:** Download MURA (40,561 images, 14,863 studies, 7 body parts), verify study-level labels (normal/abnormal), report class balance per body part (likely varies—wrist may have more abnormalities than humerus). Access FracAtlas separately, verify bounding box annotations exist, report size. **Critical:** your train/val splits must be grouped by patient ID—if a patient has multiple studies, all their images must go into the same split to prevent data leakage.
- **Resolve multi-image studies:** MURA provides multiple views per study (e.g., AP and lateral X-rays of the same wrist). Decide now: (a) treat each image independently (simpler, more training data, but ignores multi-view information), or (b) aggregate predictions per study (max-pool, average, or attention over views). Option (a) is recommended as baseline; option (b) is a good ablation.
- **Simple baseline first:** Train ResNet-50 (ImageNet pretrained, freeze layers 1–3, fine-tune layer 4 + FC head) on MURA for binary abnormality detection (ignoring body part). Use weighted BCE (inverse frequency), Adam (lr=1e-4), train 20 epochs. Report AUC, sensitivity, specificity, and Cohen’s kappa on val set. This establishes your floor before ConvNeXt.
- **Multi-task head design:** Implement the shared ConvNeXt backbone with two separate FC heads: Head A for 7-class anatomy (softmax + cross-entropy), Head B for binary abnormality (sigmoid + weighted BCE). Define loss weighting: $\mathcal{L} = \lambda_1 \mathcal{L}_{\text{anatomy}} + \lambda_2 \mathcal{L}_{\text{abnormality}}$, start with $\lambda_1 = \lambda_2 = 1.0$.
- **Augmentation design:** Contrast adjustment ($\pm 20\%$) and small rotations ($\pm 15^\circ$) are correct for X-rays. **Do not** flip—anatomical laterality matters (left vs right wrist). Do not crop aggressively—fracture lines can be subtle and near image borders.

Ablations

- **Single-task vs multi-task:** Compare ConvNeXt trained for abnormality-only vs multi-task (anatomy + abnormality). Hypothesis: multi-task improves abnormality AUC by +2–4% because anatomy classification forces anatomically meaningful features, especially for body parts with fewer training examples.
- **Per-body-part performance:** Report AUC and sensitivity *separately* for each of the 7 body parts. In clinical settings, also report Positive Predictive Value $PPV = TP / (TP + FP)$ per body part—PPV is what clinicians care about (“given a positive prediction, how likely is it real?”). Hypothesis: wrist and finger (common fracture sites) have higher AUC than humerus and shoulder (fewer examples, more complex anatomy). Plan a breakdown of F1 by anatomy type to reveal where the model fails most.
- **ConvNeXt vs ResNet-50:** Compare both backbones with identical training setup. Hypothesis: ConvNeXt improves AUC by +1–3% due to larger receptive field (7×7 kernels) capturing subtle fracture lines that 3×3 convolutions miss.

- **Cross-dataset generalization (FracAtlas):** Evaluate MURA-trained model on FracAtlas (classification only, ignore bounding boxes). Report $\Delta_{\text{gen}} = \text{AUC}_{\text{MURA val}} - \text{AUC}_{\text{FracAtlas}}$ to measure domain shift from MURA’s diverse imaging conditions to FracAtlas.

Risks & mitigations

- **Risk:** Multi-task loss weighting (λ_1, λ_2) is sensitive—anatomy classification is a much “easier” task than fracture detection, so the anatomy loss may dominate the gradient, causing abnormality detection to stall—**Mitigation:** Tune λ on validation set. You will likely need to heavily down-weight anatomy: try $\lambda_1 \in \{0.1, 0.3, 0.5\}$ with $\lambda_2 = 1.0$. Alternatively, use GradNorm or uncertainty-based task weighting if manual tuning fails.
- **Risk:** MURA labels are study-level (not image-level), so if you train image-level, some individual views may be mislabeled (e.g., normal lateral view in an abnormal study where only the AP view shows pathology)—**Mitigation:** Acknowledge this noise, validate with study-level aggregation (predict abnormal if *any* view is predicted abnormal), compare image-level vs study-level AUC.
- **Risk:** Proposal references YOLO and object detection metrics but MURA has no bounding boxes—**Mitigation:** Remove YOLO and localization references from your methodology. Focus on classification. If you want localization, use Grad-CAM to highlight where the model “looks” (qualitative, not quantitative localization).

Open questions

- Will you use ConvNeXt-Tiny (28M params), ConvNeXt-Small (50M), or ConvNeXt-Base (89M)? Tiny is recommended for comparability with ResNet-50 (25M). Specify before coding.
- How will you aggregate multi-view predictions per study—max-pool (predict abnormal if any view exceeds threshold), average probability, or learned attention over views? Define before Milestone-2.
- The proposal mentions one team member (Srivatsa) is not listed in the Individual Contributions section—clarify team member roles and update the contributions section.